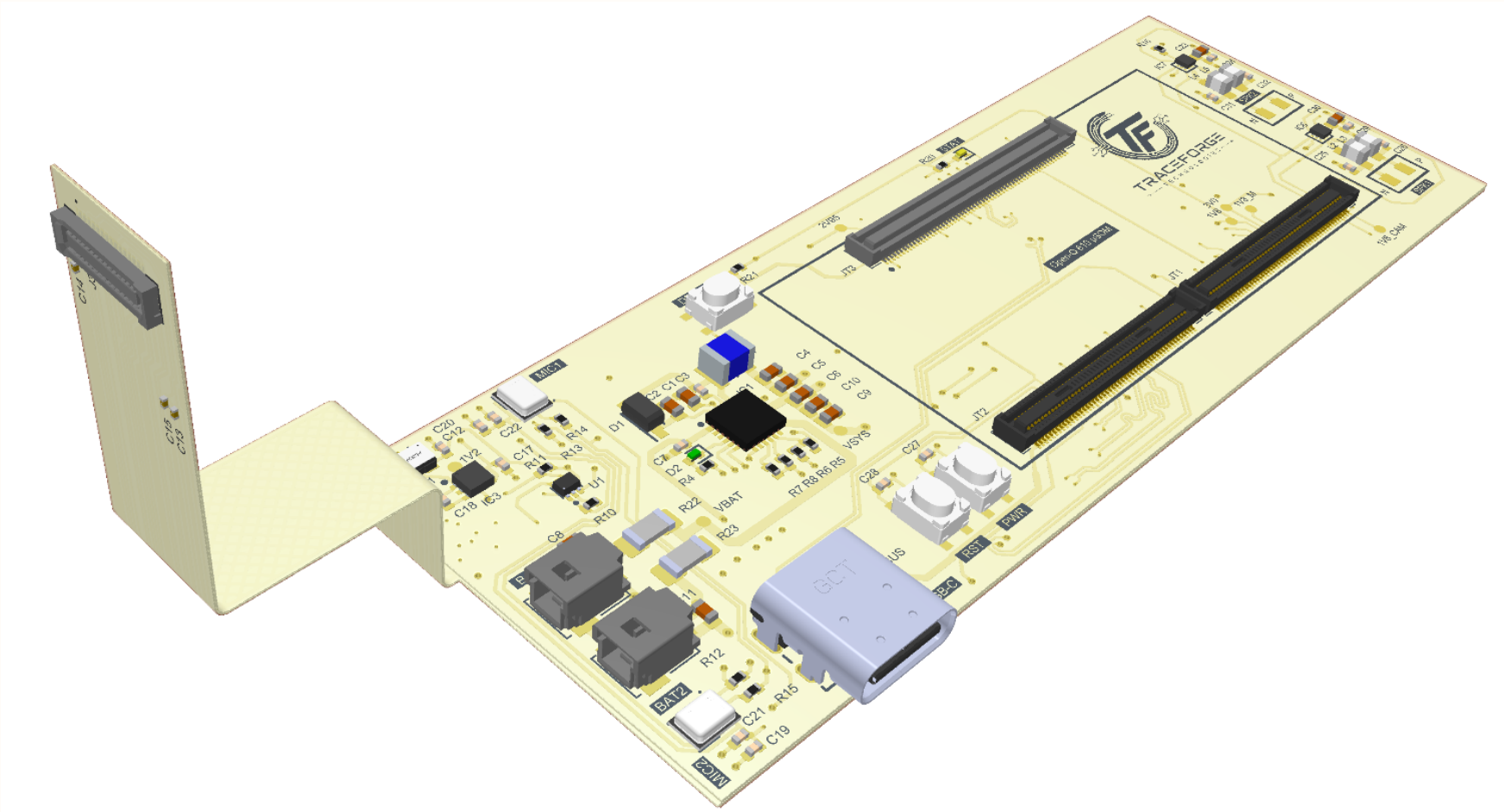


AI Powered Smart Glasses

Sheet Details
[1] Title Page
[2] Open-Q 610 μ SOM
[3] Power Section
[4] Camera
[5] Audio
[6] IO Peripherals



Project Overview

Processing Unit:

- QualComm Open-Q 610 SOM

Security:

- SHA-256 Cryptography

PMIC:

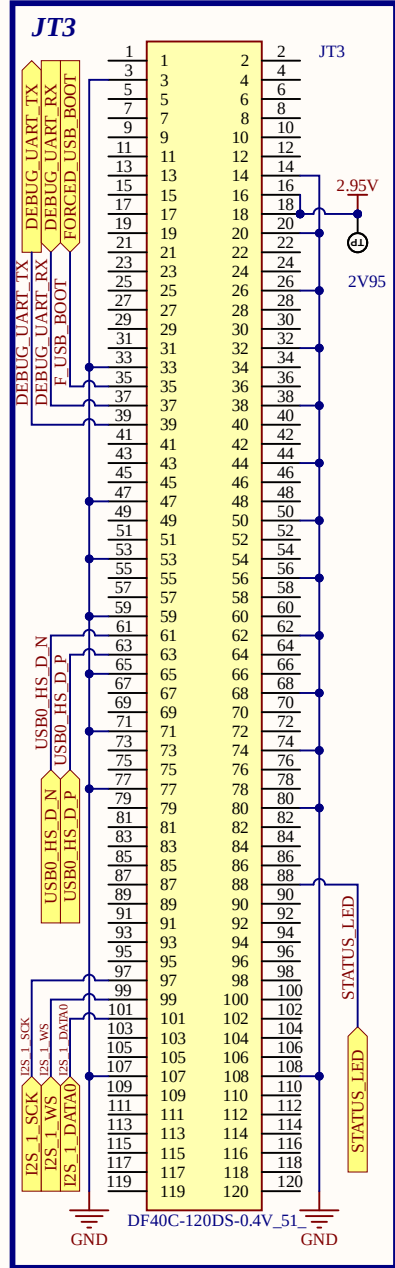
- BQ25895RTWR

Others:

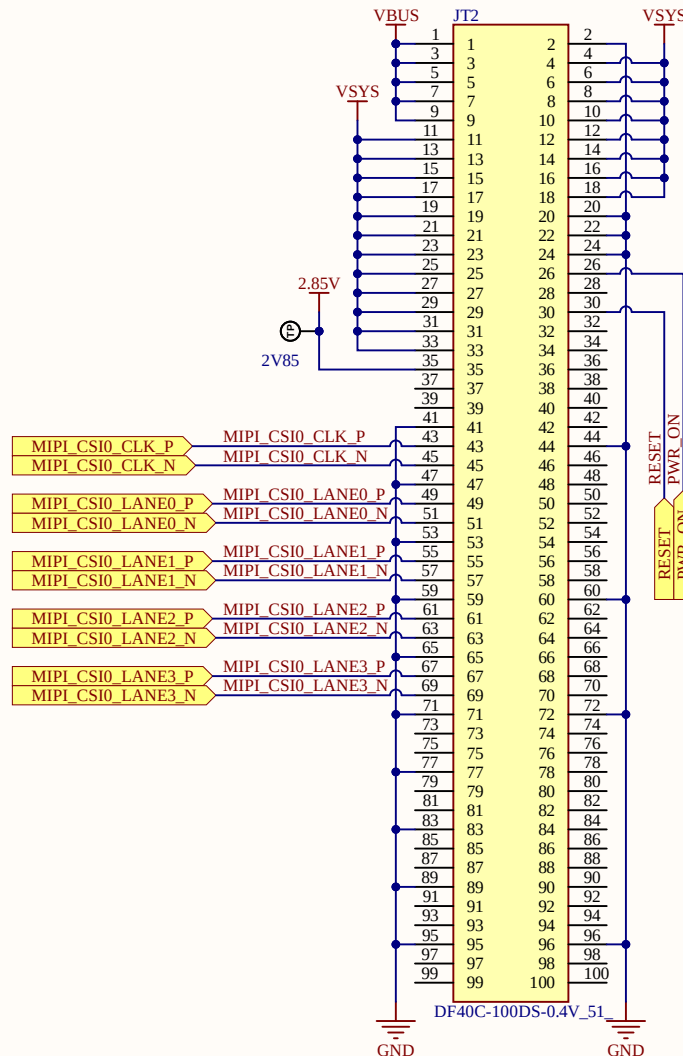
- Sony IMX219 Camera
- Micophone (Left & Right)
- 8 Ohms, 2W Speaker
- MAX98357AEWL+ Amplifer
- DC Vibration Motor

Title: AI Powered Smart Glasses			
Page Contents: Title Page.SchDoc			
Drawn By: Engr. Usman Haider	Checked By: *		
Size: A3	Company: TraceForge Technologies	REV. A	
Date: 8/30/2026	Time: 5:30:48 PM	Sheet 1 of 6	
Email: traceforgetechnologies@gmail.com			
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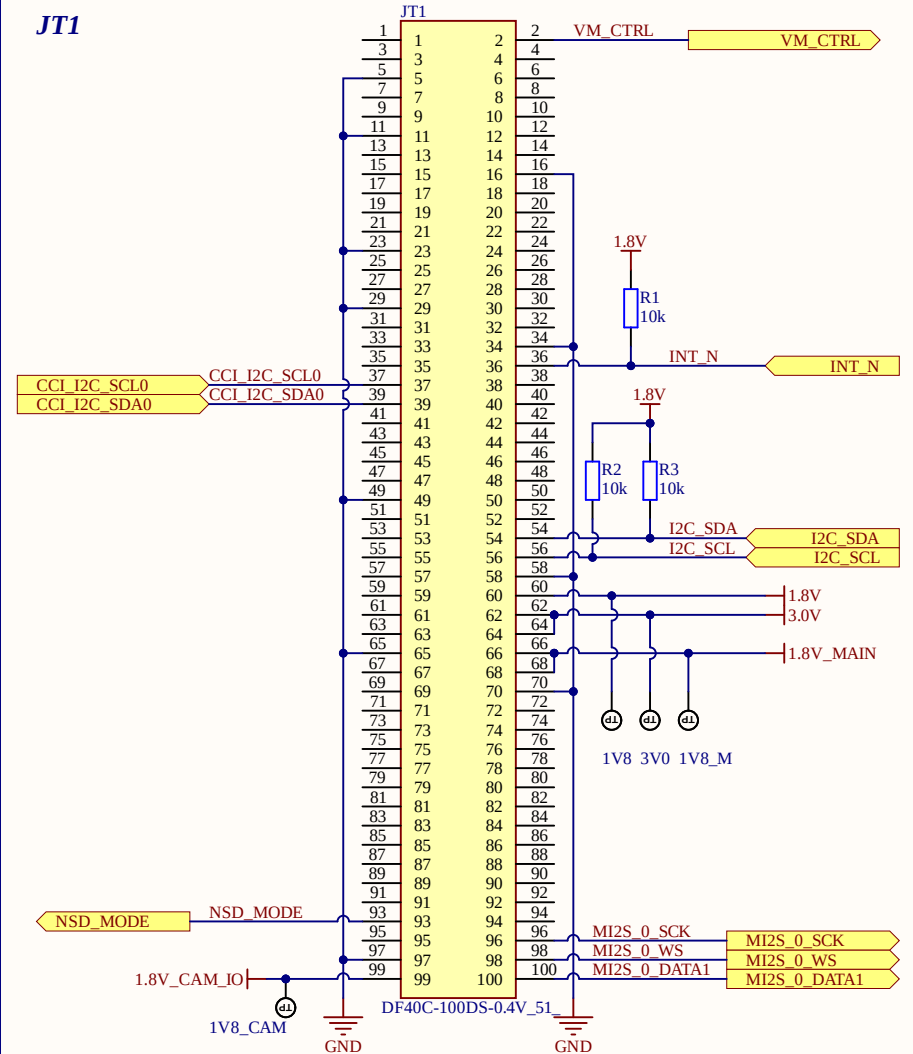
Magazine Connectors for SOM (Open-Q 610)



JT2



JT1



Title:

AI Powered Smart Glasses

Page Contents: Open-Q 610 μ SOM.SchDoc

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Checked By: *

Size: A3 Company: TraceForge Technologies

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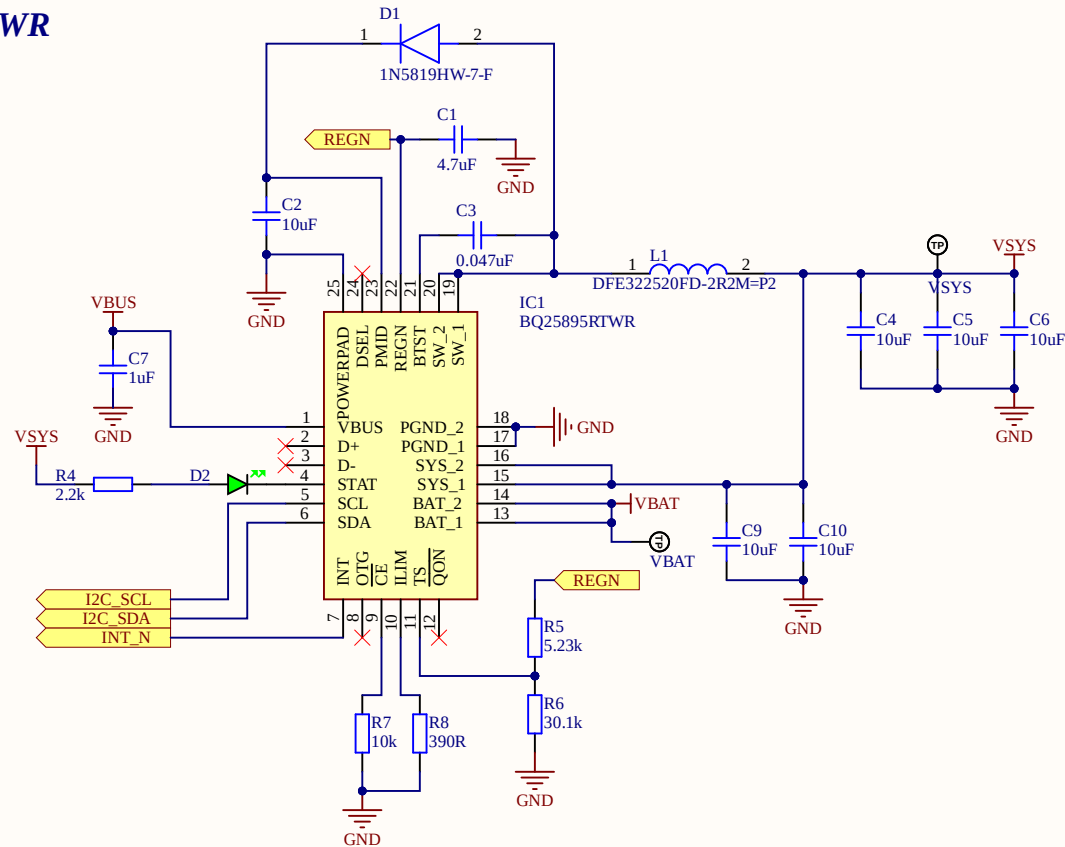
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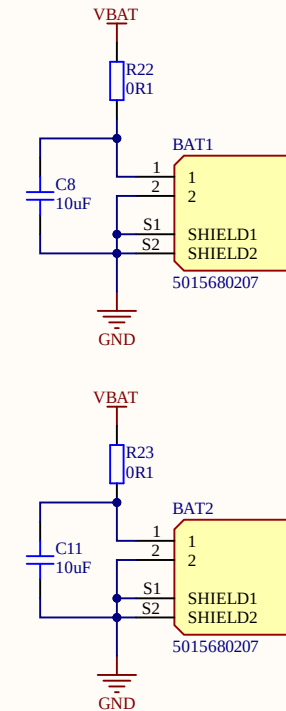
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PMIC- BQ25895RTWR

D2: Battery Status LED

2 X Li-Po Batteries



Title:

AI Powered Smart Glasses

Page Contents: [Power section.SchDoc](#)

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Size: A3	Company: <i>TraceForge Technologies</i>
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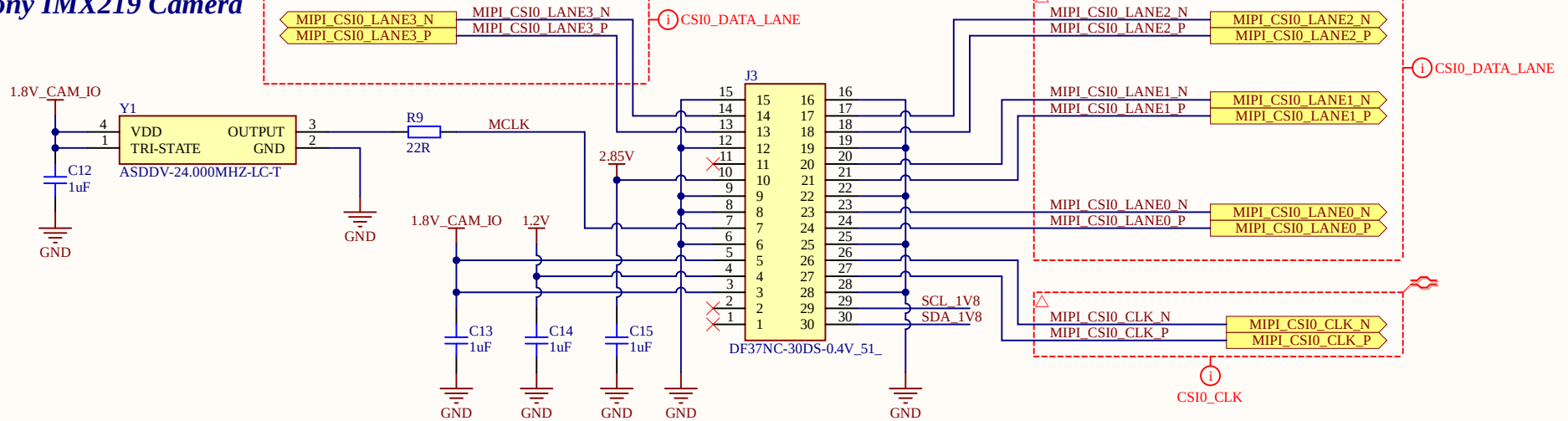
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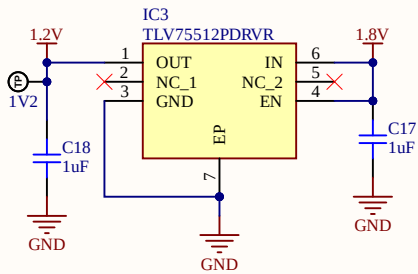
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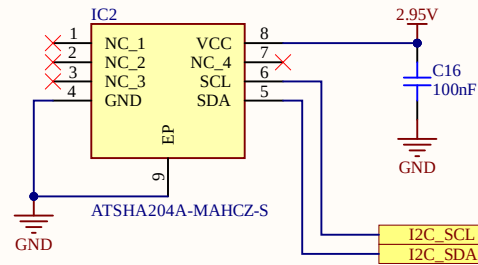
Sony IMX219 Camera



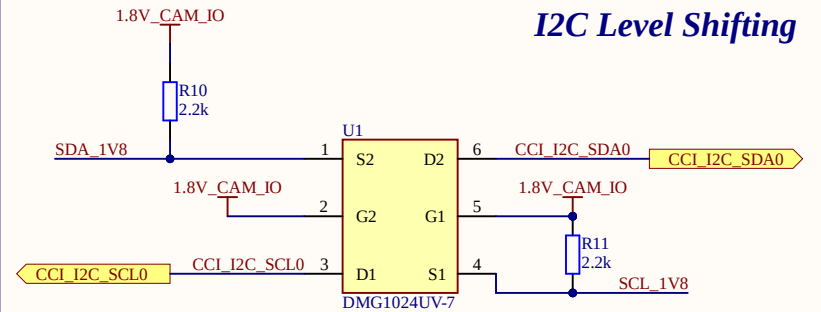
LDO-1V2



Cryptography



I2C Level Shifting



Title:

AI Powered Smart Glasses

Page Contents: Camera.SchDoc

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Size: A3 Company: TraceForge Technologies

REV. A

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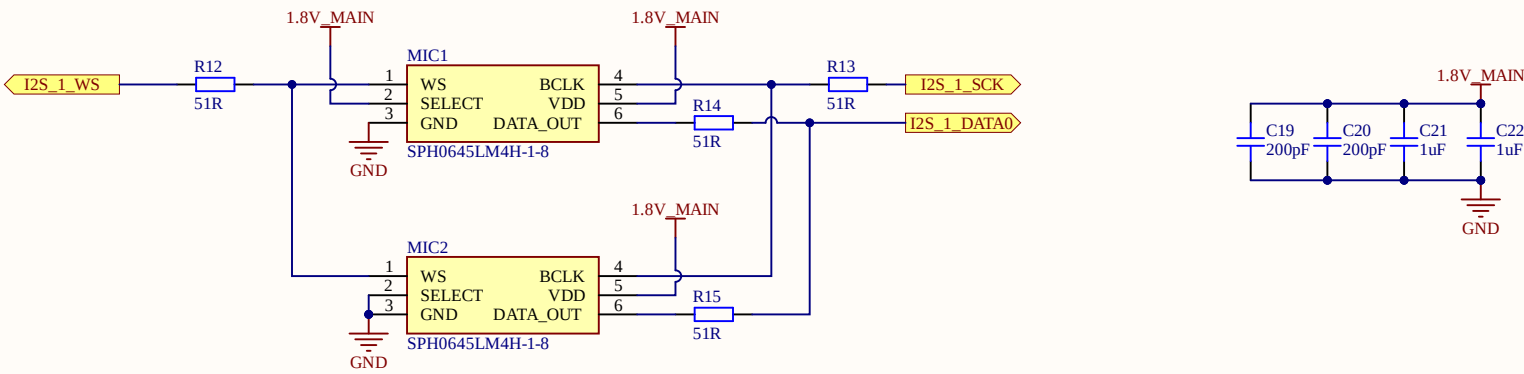
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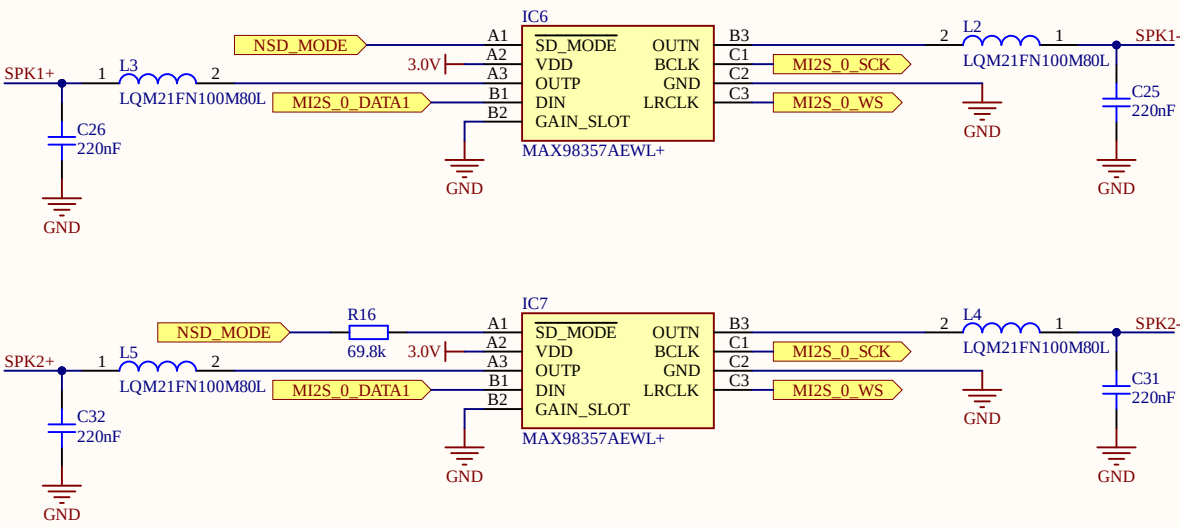
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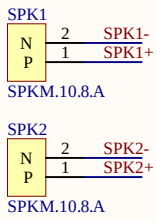
Micophone (Left & Right)



Audio Amplifiers



Speaker (Left & Right)



USB-C

The diagram illustrates the internal wiring of a USB-C connector. The connector (J4) has pins for VBUS, GND, and SHELL. The VBUS pin is connected to a 5V source and a diode D4. The GND pin is connected to ground. The SHELL pin is connected to ground. The DP1, DP2, DN1, and DN2 pins are connected to a USB0_HS_DIFF signal. The SBU1 and SBU2 pins are connected to ground. The CC1 and CC2 pins are connected to a 5.1k resistor (R18) and a 5.1k resistor (R19) respectively. The DP1 and DP2 pins are connected to a diode D5. The DN1 and DN2 pins are connected to a diode D6. The USB0_HS_DIFF signal is shown as a differential pair (USB0_HS_D_P and USB0_HS_D_N) connected to a USB-C connector.

Vibration Motor

The diagram illustrates a circuit for driving a vibration motor (M1) using an AO3407A MOSFET (Q1). The MOSFET's source (pin 3) is connected to ground. The gate (pin 1) is connected to a 2.95V source and a 10k resistor (R17), which is also connected to a control signal (VM_CTRL). The drain (pin 2) is connected to the anode of a diode (D3, 1N5819HW-7-F). The diode's cathode (pin 2) is connected to ground. The MOSFET's drain (pin 2) is also connected to the VDD pin (pin 1) of the vibration motor (M1, PJFVMA011001B-B0). The GND pin (pin 2) of the motor is connected to ground.

The diagram shows three circuit sections:

- PWR ON:** A signal source labeled "PWR_ON" is connected to a node. This node is also connected to a 100nF capacitor (C27) to ground (GND). The node then passes through a switch labeled "PWR" to another ground (GND).
- Reset:** A signal source labeled "RESET" is connected to a node. This node is also connected to a 100nF capacitor (C28) to ground (GND). The node then passes through a switch labeled "RST" to another ground (GND).
- Forced Debugging:** A 1.8V source is connected to a 10k resistor (R21). The other end of R21 is connected to a node labeled "FORCED_USB_BOOT". This node is connected to a switch labeled "DEBUG", which is then connected to ground (GND).

Forced Debugging

Debug Port

1.8V DEBUG CONN

1
2
3
4
5

DEBUG_UART_RX
DEBUG_UART_TX
RESET

GND

Debug

Title: ***AI Powered Smart Glasses***

Page Contents: [IO Peripherals.SchDoc](#)

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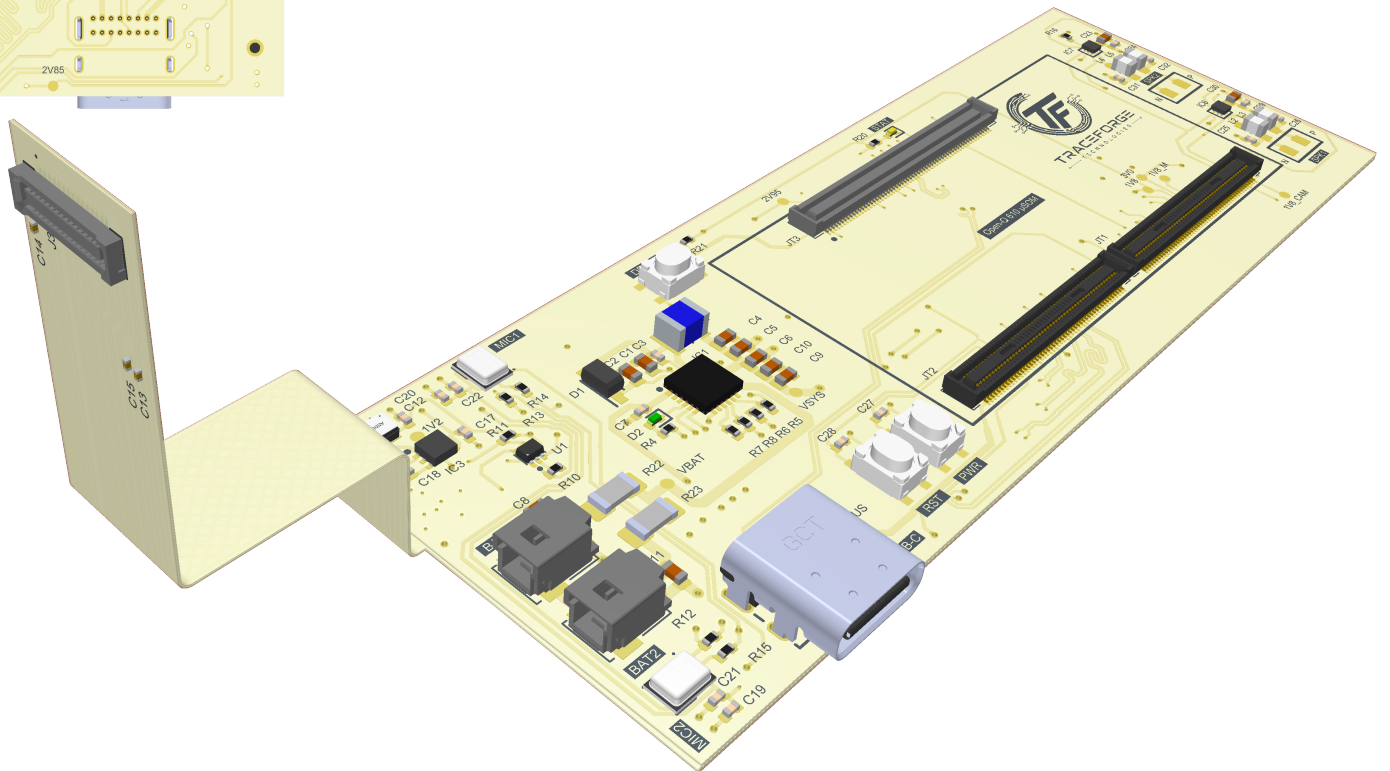
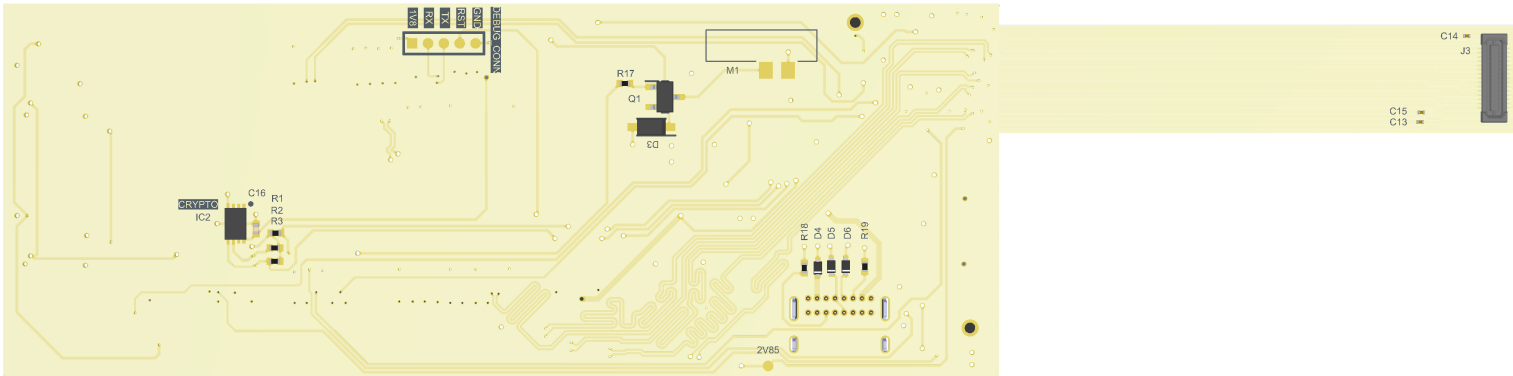
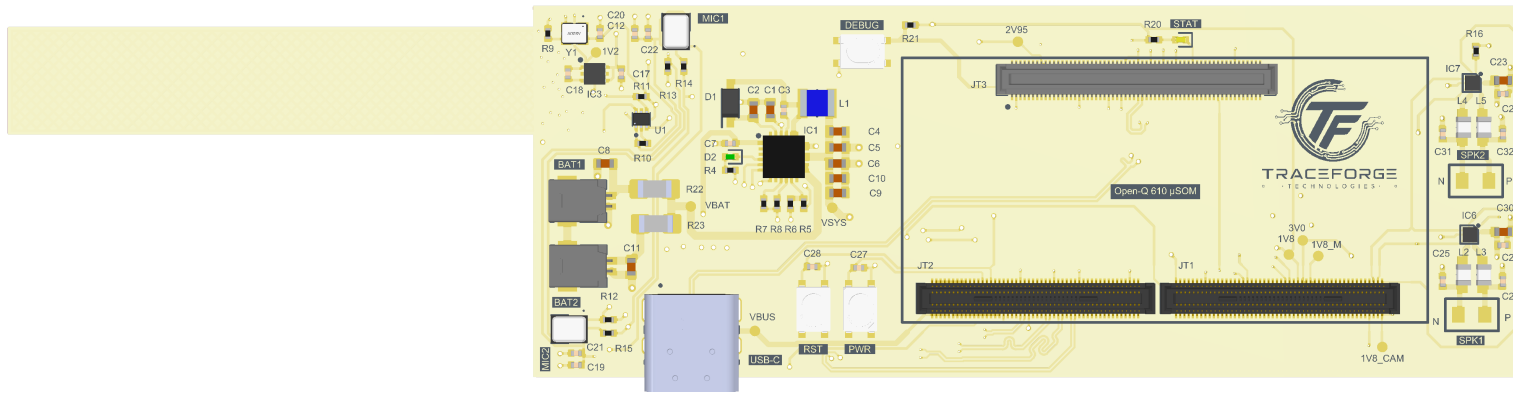
A

B

C

D

E



A

B

C

D

E

A

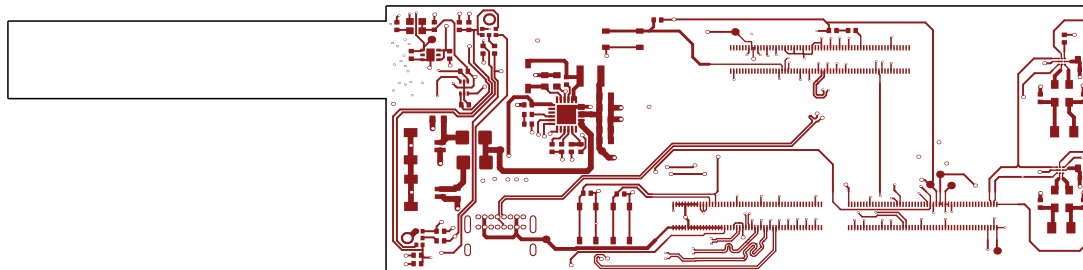
B

C

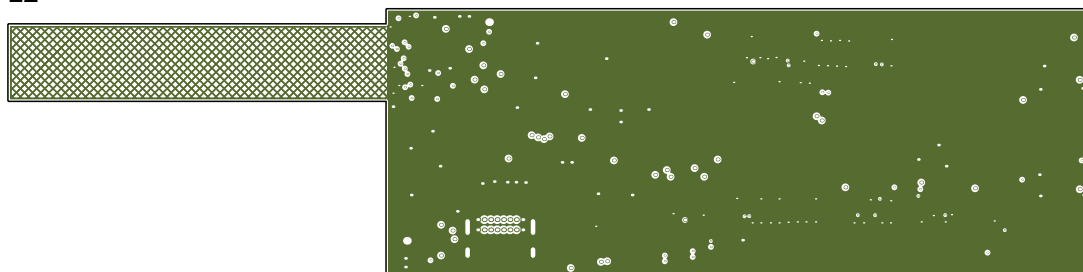
D

E

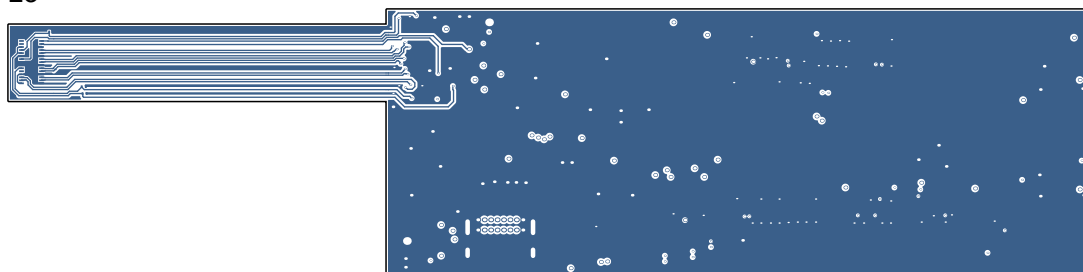
L1



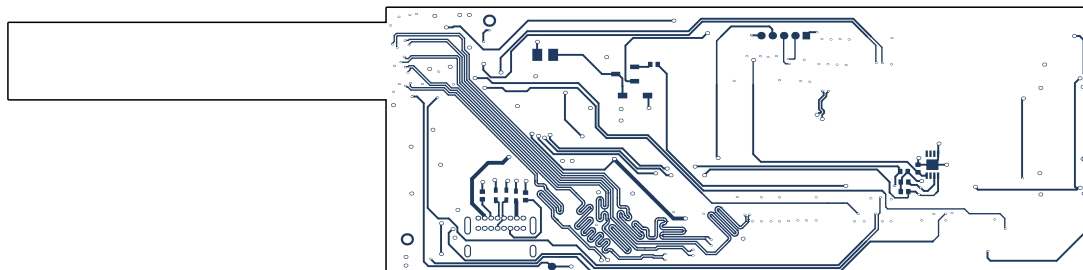
L2



L3



L4



A

B

C

D

E

Layer Stack Legend

	Material	Layer	Thickness	Dielectric Material	Type	Gerber
		Top Overlay			Legend	GTO
	Surface Material	Top Solder	0.03mm	SM-001	Solder Mask	GTS
	Copper	L1	0.04mm		Signal	GTL
	Core		0.20mm	Core-024	Dielectric	
	PP-004	Adhesive 1	0.07mm		Adhesive	
	CF-004	L2	0.02mm		Signal	G1
	Core		0.03mm	Core-028	Dielectric	
	CF-004	L3	0.02mm		Signal	G2
	PP-004	Adhesive 2	0.07mm		Adhesive	
	Core		0.20mm	Core-024	Dielectric	
	Copper	L4	0.04mm		Signal	GBL
	Surface Material	Bottom Solder	0.03mm	SM-001	Solder Mask	GBS
		Bottom Overlay	0.01mm		Legend	GBO
Total thickness: 0.73mm						

